

Predicting YouTube Sports Video Popularity

A Mini Data Science Project Proposal

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1. Introduction

In today's digital landscape, YouTube has evolved beyond an entertainment platform into a powerful socioeconomic ecosystem. With over millions of videos viewed, content visibility is increasingly determined by algorithmic factors rather than content quality alone. This project aims to identify the key factors that predict YouTube video popularity, specifically in the field of sports content, by handling and analyzing real-world datasets to uncover valuable patterns that can inform decision-making and drive actionable outcomes for content creators and digital marketers.

2. Objectives

- Fetch, explore, and preprocess YouTube sports video data to ensure data quality and readiness for analysis.
- Conduct exploratory data analysis to uncover patterns and distributions in video attributes such as subscriber count, publish time, engagement metrics, tag count, and video duration.
- Visualize key findings through correlation heatmaps, distribution plots, and trend graphs to communicate insights effectively.
- Apply Lasso Regression to identify which features are the strongest predictors of video popularity (measured by view count) while eliminating irrelevant variables through regularization.
- Provide clear, actionable recommendations for content creators on how to optimize their videos for maximum reach and engagement.

3. Data Collection

The dataset will be obtained through web scraping from YouTube using the YouTube Data API v3. The collected data will encompass the following variables for each video:

- Subscriber count: the number of subscribers of the publishing channel
- Publish time: the date and time the video was published
- Engagement metrics: like ratio (likes/views) and comment ratio (comments/views)
- Keywords count: the total number of tags/keywords associated with the video
- Video duration: the length of the video in seconds

4. Deliverables:

A Jupyter Notebook file that includes:

- Complete code for data fetching, preprocessing, analysis, and modeling.
- Visualizations and graphs presenting key patterns and findings.
- Lasso Regression model with evaluation metrics (R^2 , RMSE).
- A comprehensive summary of insights and actionable recommendations derived from the analysis.

5. Conclusion

By systematically handling and analyzing YouTube data through all phases from data fetching to machine learning modeling, this project will deliver meaningful insights into what drives sports video popularity to make more informed, data-driven decisions.